PARISHRAM 2.0 2026 (Class 12th)

Mathematics

DPP: 04

Relations and Functions

Q1 Complete set of domain of $f\left(x\right)=\frac{x-3}{x^2-4}$ is:

(A) $x\in(-2,2)\cup(3,\infty)$

(B) $x\in(-\infty,-2)\cup(2,3]$

(C) $x\in(-2,2)\cup[3,\infty)$

(D) $x\in R-\{-2,2\}$

Q2 If $f\left(x\right)=\frac{x-10}{10-x}$, then domain of $f(x)$ is

(A) $\{5\}$ (B) $R-\{10\}$

(C) R (D) None of these

Q3 The range of the real function $f\left(x\right)=\frac{x+1}{x-3}$ is

(A) $R-\{3\}$

(B) $R-\{1\}$

(C) R

(D) $R-\{-3\}$

Q4 The range of the function $f\left(x\right)=\frac{1+x^2}{x^2}$ is equal to

(A) $(0,1)$ (B) $[0,1]$

(C) $(1,\infty)$ (D) $[1,\infty)$

Q5 Let f : [2, ∞) → R be the function defined by $f(x) = x^2 - 4x + 5$. Then the range of f is

(A) R (B) $[1,\infty]$

(C) $[4,\infty)$ (D) $[5,\infty)$

Q6 Find the domain and range of a function $f(x) = 3x^2 - 5$.

(A) Domain $=(-\infty,\infty)$, Range $=[-5,\infty)$

(B) Domain $=(-\infty,\infty)$, Range $=(5,\infty)$

(C) Domain $=(-\infty,\infty)$, Range $=(-\infty,\infty)$

(D) Domain $=(-\infty,\infty)$, Range $=(-\infty,5]$

Answer Key

Q1	D
Q2	B
Q3	B
Q4	C
Q5	B
Q6	A

Hints & Solutions

Q1 **Text Solution:**

For $f(x)$ to be defined

$x^2 - 4 \neq 0$

$\Rightarrow x^2 \neq 4$

$\Rightarrow x \neq \pm\sqrt{4}$

$\Rightarrow x \neq \pm 2$

Video Solution:

Q2 **Text Solution:**

For $f(x)$ to be defined, $10 - x \neq 0$

$\Rightarrow x \neq 10$

$\therefore$ Domain $= R\text{~}\{10\}$

Video Solution:

Q3 **Text Solution:**

Domain

For $f(x)$ to be defined

$x - 3 \neq 0$

$\Rightarrow x \neq 3$

$\therefore$ Domain = $R\text{~}\{3\}$

Range

Let $y = f(x)$

$\Rightarrow y = \frac{x+1}{x-3}$

$\Rightarrow xy - 3y = x + 1$

$\Rightarrow xy - x = 3y + 1$

$\Rightarrow x(y - 1) = 3y + 1$

$\Rightarrow x = \frac{3y+1}{y-1}$

$\Rightarrow y \in R\text{~}\{1\}$

$\therefore$ Range $= R\text{~}\{1\}$

Video Solution:

Q4 **Text Solution:**

Domain

For $f(x)$ to be defined

$x^2 \neq 0$

$\Rightarrow x \neq 0$

$\therefore$ Domain $= R\text{~}\{0\}$

Range

$f\left(x\right) = \frac{1+x^2}{x^2}$

$f\left(x\right) = \frac{1}{x^2} + 1$

Now $x^2 \geqslant 0$ for all $x \in R$

$\Rightarrow \frac{1}{x^2} > 0$

$\Rightarrow 1 + \frac{1}{x^2} > 1$

$\Rightarrow f(x) > 1$

$\Rightarrow$ Range of $f(x) = (1, \infty)$

Video Solution:

Q5 **Text Solution:**

We have, $f(x) = x^2 - 4x + 5 = (x - 2)^2 + 1$

Now, $x \in [2, \infty) \Rightarrow x - 2 \geq 0 \Rightarrow (x - 2)^2 + 1 \geq 1$

$\Rightarrow f(x) \geq 1 \Rightarrow f(x) \in [1, \infty)$

Hence, range $(f) = [1, \infty)$.

Video Solution:

Q6 Text Solution:

We know that the domain of a function is the set of input values for f, in which the function is real and defined.

The given function has no undefined values of x.

Thus, for the given function, the domain is the set of all real numbers.

Domain $= [-\infty, \infty]$

Also, the range of a function comprises the set of values of a dependent variable for which the given function is defined.

Ley $y = 3x^2 - 5$

$3x^2 = y + 5$

$x^2 = (y + 5)/3$

$x = \sqrt{(y+5)/3}$

Square root function is defined for non-negative values.

So, $[(y + 5)/3] \geq 0$

This is possible when $(y + 5)/3 \geq 0 \Rightarrow y \geq -5$.

Hence, the range of f(x) is $[-5, \infty)$.

Video Solution:

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